



SUJET STATAPP 356 - 2026-2027

Sujet proposé par : Molinaro Matéo - mateo.molinaro@via-am.com

Organisme : VIA AM

Catégorie : Finance

Sujet :

Finding Factors Using NLP/LLMs on 10-K/10-Q/8-K Filings

Description du sujet :

Problem statement

Annual and quarterly SEC filings (10-K, 10-Q, and the 8-K filings that accompany earnings announcements) contain rich, mostly unstructured information ? risk factors, Management Discussion & Analysis (MD&A), footnotes ? that is only partly captured by traditional structured factors built from accounting and price data. The project would use NLP/LLM techniques (bag-of-words/TF-IDF baselines, embeddings, fine-tuned language models, or prompted large language models) to extract candidate factors from filing text ? e.g. year-over-year change in risk-factor language, tone, topic exposure ? and test whether these text-based signals add incremental predictive power for the cross-section of stock returns, on top of standard factor libraries.

The project follows a natural three-step progression, taking the text-based factor(s) from construction all the way to a portfolio-level assessment:

1. Standalone factor construction and evaluation.

Build one or several text-based factors

(e.g. textual similarity/change scores, dictionary-based tone, topic exposure via embeddings or an LLM) and assess their standalone predictive power using standard cross-sectional asset-pricing tools: information coefficient, decile/quintile portfolio spreads, factor Sharpe ratio, and spanning tests against existing factors.

2. Incremental value inside a return-prediction ML model. Feed the text-based factor(s) as additional inputs into a standard ML return-prediction pipeline (e.g. gradient boosting, elastic net, or a simple neural network) alongside a broad panel of standard equity factors (the ?factor zoo?), and measure incremental predictive contribution via feature importance, SHAP values, and out-of-sample ablation (with vs. without the text factor(s)).

3. Integration into a multifactor equity portfolio.

Combine the text-based factor(s) with existing style factors (value, momentum, quality, low-volatility) inside a multifactor portfolio-construction framework, and evaluate the marginal contribution to risk-adjusted performance, turnover, capacity, and correlation with existing factor exposures.

Time permitting, the project could be extended with a fourth, more exploratory angle: an event-based quant strategy, using the text-based signal(s) as a trigger or conditioning variable for an event-driven strategy around earnings announcements or new filings (in the spirit of post-earnings-announcement drift), testing whether text-derived information is impounded into prices with a lag that a systematic strategy could exploit.

Données du sujet :

Possible data

SEC EDGAR full-text filings (10-K, 10-Q, 8-K), e.g. via the EDGAR full-text search API or a commercial wrapper (sec-api.io); Open Source Asset Pricing (<https://www.openassetpricing.com/>) for

the benchmark equity factor library used in angles 2?3; standard CRSP/Compustat-style price and returns data (or a comparable open alternative) for return construction and event studies.

Modèles envisagés du sujet :

ML (lasso, ridge, elastic net, lightgbm, xgboost) / NLP (tf-idf, transformers) / LLM (finBERT) / econometrics (linear regression)

Bibliographie du sujet :

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Notation du sujet :

Commentaires :